

The impact of social networking sites on the students around Kolkata

Sourav Dey, Manish Chandra Roy, Swarnendu Dey, Roshni Majhi

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I think you know this icons very well!!

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Introduction

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- The prevalence of social networking sites (SNS) has significantly increased over the past decade, reshaping how students interact and communicate. The rise of platforms like Facebook, Twitter, and Instagram has revolutionized the way students connect with peers, share information, and express themselves online.

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- The prevalence of social networking sites (SNS) has significantly increased over the past decade, reshaping how students interact and communicate. The rise of platforms like Facebook, Twitter, and Instagram has revolutionized the way students connect with peers, share information, and express themselves online.
- The popularity of social networking sites began to soar around the mid-2000s, with Facebook emerging as a dominant player in the market. Over the years, the landscape has evolved, with new platforms and features continuously being introduced. This rapid growth has had a profound impact on student life, influencing their social interactions, academic pursuits, and overall well-being.

Introduction (Contd.)

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- The outbreak of the COVID-19 pandemic further amplified the role of social networking sites in students' lives, as quarantine and social distancing measures forced them to rely more on these platforms for socializing and staying connected with friends and classmates. This increased reliance on social networking sites has led to changes in how students manage their time, with many spending more hours online than before.

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- As a result, we have witnessed a significant shift in student's social habits and behaviors, with social networking sites playing a central role in shaping their identities and relationships. Understanding these changes is crucial for educators, parents, and policymakers to support students in navigating the digital landscape and ensuring a healthy balance between online and offline interactions.

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- 1 Do the majority of students have a high daily screen time?

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- 1 Do the majority of students have a high daily screen time?
- 2 Does the different types of sites/apps used by students affect daily screen time significantly?

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- ① Do the majority of students have a high daily screen time?
- ② Does the different types of sites/apps used by students affect daily screen time significantly?
- ③ Do students have a greater preference of using social media for entertainment purposes than educational?

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- ④ Does social media addiction carry a significant negative influence on academic performances of students according to them?

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- 1 Do the majority of students have a high daily screen time?
- 2 Does the different types of sites/apps used by students affect daily screen time significantly?
- 3 Do students have a greater preference of using social media for entertainment purposes than educational?
- 4 Does social media addiction carry a significant negative influence on academic performances of students according to them?
- 5 Those who prefer to use social media frequently, what kind of impacts are visible on them?

Data Collection Methods

Plan of enquiry

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- Here, as our survey method we have chosen online questionnaire survey because of the benefits.
- It saves our time and resources.
- High reach of circulation unlike door to door survey.
- Although we have some cons in questionnaire like Our reach is restricted to only those who use mobile and have active internet connection.

Target Population

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Our target population was the people living in Kolkata and its surrounding places and also villages, as we are interested to know the change in their internet usage habits in recent times.

Design of the Questionnaire

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In order to frame a proper questionnaire we started with the basic questions which we thought will be required in our analysis and our first set of questions was :

Topic

The impact of social networking sites on the studies of students near Kolkata

Provisional Questionnaire:

Dear respondent,

We are students of Presidency University working on a survey report about the "The impact of social networking sites on the studies of students near Kolkata". Please fill up the questionnaire the information given by you will be kept confidential and used for research purpose only.

6. How often do you use social networking sites?

- i) Not at all
- ii) Rarely
- iii) Infrequently
- iv) Frequently
- v) Always

7. How long hours a day do you spend on social networking websites?

- i) 0-1 hours
- ii) 1-2 hours
- iii) 2-3 hours
- iv) 3-5 hours
- v) 5+ hours

8. If you are college student which stream you're in?

- i) Sc
- ii) Arts

9. Do you like to read books as daily basis?

- i) Yes
- ii) No

Part-A

1. Name:

2. Age:

3. Sex:

4. Educational Qualification

- i) UG
- ii) PG
- iii) Graduated
- iv) Profession

5. Do you use social networking sites?

- i) Yes
- ii) No

[If the answer of the above question is "No". Then you don't need to fill the following!]

10. Daily data usage: (*in Giga Bytes*)

- i) 1-1.5
- ii) 1.5-2.5
- iii) 2.5-5
- iv) 5+

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11. Favorite Social networking platform to spend time:

- i) Instagram
- ii) Facebook
- iii) YT
- iv) Chrome
- v) Telegram
- vi) X
- vii) Others

12. Why do you spend time on social networking sites?

- i) Chatting & relaxing
- ii) For study
- iii) Writing blogs
- iv) Others

13. What do you prefer? Which one do you prefer?

- i) Offline classes
- ii) Online classes

14. Do you feel stressed out for over usage of social networking sites.

- i) Yes
- ii) No

15. Does this effect your life?

- i) Yes
- ii) No

16. Money spends on the internet in each month: (*In rupees*)

- i) 0-250
- ii) 250-500
- iii) 500-1,000
- iv) 1000+

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That was just an overview to frame our first questionnaire. In our first questionnaire the errors which we have done were as follows:

- The order of the questions were not in a proper manner which may disturb the student and he/she may not complete the form.

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- i) Yes
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That was just an overview to frame our first questionnaire. In our first questionnaire the errors which we have done were as follows:

- The order of the questions were not in a proper manner which may disturb the student and he/she may not complete the form.
- Few questions that were not necessary were also included in the questionnaire which unnecessarily increased the length of the questionnaire such as :

15. Does this effect your life?

- i) Yes
- ii) No

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Those questions were not useful in our analysis so we removed them and replaced them by useful questions in our final questionnaire.

- Lastly, there were few mistakes in the options and in framing the question properly.

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- Later we have done the corrections in our first questionnaire.

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Those questions were not useful in our analysis so we removed them and replaced them by useful questions in our final questionnaire.

- Lastly, there were few mistakes in the options and in framing the question properly.
- Later we have done the corrections in our first questionnaire.
- Our final questionnaire is shown in the next slide

Contd.

The impact of social networking sites on the students around Kolkata

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Name: * Your answer	
Sex: * ☐ Male ☐ Female ☐ Prefer not to say ☐ Other: _____	Favorite social networking platform to spend time on: * ☐ whatsapp ☐ Instagram ☐ Facebook ☐ YouTube ☐ Telegram ☐ twitter ☐ None ☐ Other: _____
Age: * Your answer	What is the significant purpose of yours to use social networking sites? * ☐ Writing blogs ☐ Gathering information ☐ Studying ☐ Chaffing and relaxing ☐ Other: _____
Annual family income : * ☐ below 60000 ☐ 60,000 - 3,40,000 ☐ 3,40,000 - 7,80,000 ☐ 7,80,000 - 10,00,000 ☐ Above 10,00,000	From where do you get to know the socio- political scenario of recent times? * ☐ Newspapers ☐ news channels ☐ Social networking sites ☐ Not interested at all ☐ Other: _____
Where do you stay? * ☐ Kolkata ☐ Surroundings ☐ Villages ☐ Other: _____	

Final Questionnaire (Contd.)

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Do you feel stressed out for using too much of social networking sites? *

- ☐ Yes
☐ No

In your opinion how would you rate your academic satisfaction out of 5. *

- ☐ 1
☐ 2
☐ 3
☐ 4
☐ 5
☐ Not at all

In your opinion how would you rate your professional satisfaction out of 5. professional satisfaction *

- ☐ 1
☐ 2
☐ 3
☐ 4
☐ 5
☐ Not at all

How would you find yourself in a social gathering? *

- ☐ Extroverted
☐ Introverted

Data description

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- We have got 150 observations with 22 variables where 17 variables are categorical and 5 are numerical.
- We got the following variables w.r.t. Q1,Q2, and so on respectively:
 - Q1: Time
 - Q2: Name
 - Q3: Gender
 - Q4: Age
 - Q5: Family income
 - Q6: Locality
 - Q7: Commute hours to go to their college or university
 - Q8: How often do they use social networking sites
 - Q9: Time spending on the internet
 - Q10: Academic Stream
 - Q11: Do they enjoy reading?
 - Q12: Language they prefer for reading

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- Q13: Favorite genre
- Q14: Favorite platform
- Q15: Significant purpose of yours to use social networking sites
- Q16: Where do you get to know the socio - political scenario
- Q17: Subscription of any online education course
- Q18: Do you feel stressed out for using too much of social networking sites
- Q19: Academic satisfaction out of 5
- Q20: Professional satisfaction out of 5
- Q21: How would you find yourself in a social gathering
- Q22: Sleep hours

Statistical methodology

Scrutiny of Data

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After collecting the data we found a lot of information which were incomplete in the disguise of missing. These data were eliminated after complete case analysis, otherwise these could've affected the summary of the variable behaving as outliers. Finally after scrutinization we have 140 observations.

Statistical methodology (Contd.)

Variable pre processing

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Many of the respondents answered the Question: “How many hours a day do you spend time on social networking websites on an average?” as a range like 3-4, 5-6 etc. As we asked value as an average, we replaced the given range as the mean of the two values given.

Exploratory Data Analysis

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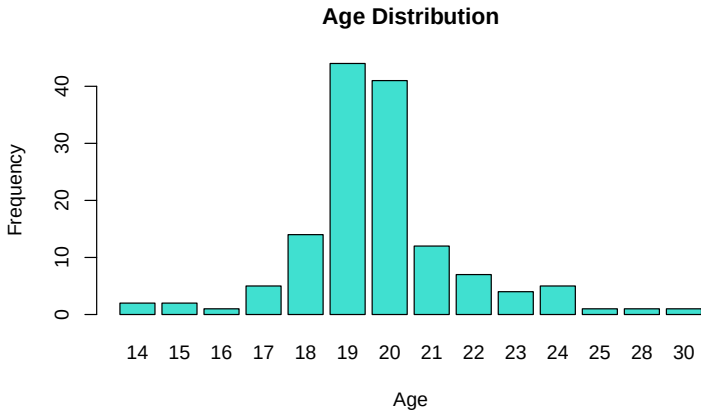
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- We start by showing the Column diagram (bar plot) of the age distribution & pie chart for annual family income from where we were collected the data.

```
'data.frame': 140 obs. of 22 variables:
 $ Q1 : chr "2024/04/09 6:24:48 pm GMT+5:30" "2024/04/09 6:33:21 pm GMT+5:30" "2024/04/09 6:52:
 $ Q2 : chr "Sangbartta Banerjee " "Pratik Majumdar " "Upasak Chatterjee " "Sneha Kundu " ...
 $ Q3 : chr "Male" "Male" "High-Elf" "Female" ...
 $ Q4 : int 19 19 20 19 20 21 19 24 20 20 ...
 $ Q5 : chr "60,000 - 3,40,000" "60,000 - 3,40,000" "Below 60000" "3,40,000 - 7,80,000" ...
 $ Q6 : chr "Surroundings" "Kolkata" "Kolkata" "Kolkata" ...
 $ Q7 : chr "1.5" "1" "less than 30 minutes " "less than 30 minutes " ...
 $ Q8 : chr "Frequently" "Moderately" "Not at all" "Frequently" ...
 $ Q9 : num 3.5 2 5 3 3 2 4 6 6 3 ...
 $ Q10: chr "Science" "Science" "Science" "Science" ...
 $ Q11: chr "Yes" "Yes" "No" "Yes" ...
 $ Q12: chr "English" "English" "Python" "English" ...
 $ Q13: chr "Narrative" "Sf" "Cooking" "Mystery" ...
 $ Q14: chr "YouTube" "Others" "Others" "Instagram" ...
 $ Q15: chr "Chatting and relaxing" "Chatting and relaxing" "Others" "Chatting and relaxing" ...
 $ Q16: chr "Social networking sites" "Social networking sites" "Not interested at all" "Social
 $ Q17: chr "No" "No" "Yes" "No" ...
 $ Q18: chr "Yes" "Yes" "No" "Yes" ...
 $ Q19: int 4 4 0 3 3 0 3 0 3 3 ...
 $ Q20: int 4 4 5 3 3 0 4 0 3 3 ...
 $ Q21: chr "Introverted" "Introverted" "Introverted" "Extroverted" ...
 $ Q22: num 7 7 6 6 7 5 6 6 6 9 ...
```

Demographics

- Age and Gender distribution



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- This histogram shows that our sample is largely composed of individuals aged 18 to 21, with the peak frequencies at ages 19 and 20. This demographic concentration is significant because it suggests our findings will primarily reflect the behaviors and attitudes of late teenagers and young adults.

Demographics(Contd.)

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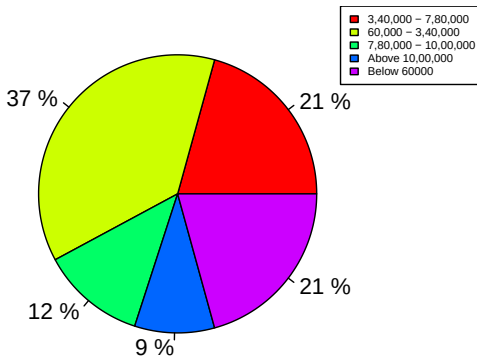
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- The smaller number of respondents in the younger and older age brackets indicates that our data may be less representative of those age groups. However, the central cluster provides a robust basis for understanding the impact of social networking sites on typical college-age students.

Demographics(Contd.)

- Family Income Distribution

Annual Family Income Distribution



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- This pie chart illustrates that a significant portion of our respondents, 37%, fall within the ₹60,000 – ₹3,40,000 income range. This is the most common income bracket among our participants.

Demographics(Contd.)

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- Approximately 21% of respondents have a family income below ₹60,000 and another 21% fall in the ₹3,40,000 – ₹7,80,000 range. These two middle-income brackets together account for a substantial portion of our sample.

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- Respondents with an annual family income above ₹10,00,000 constitute 9% of the sample, while those with an income below ₹7,80,000 - ₹10,00,000 make up the smallest group at 12%.

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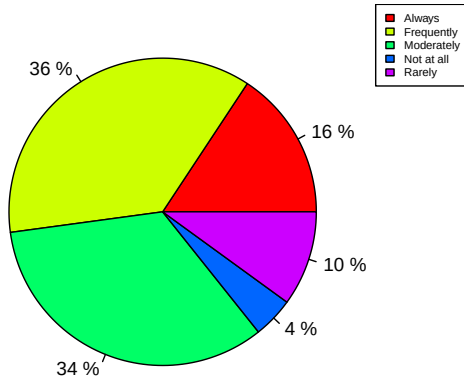
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- Respondents with an annual family income above ₹10,00,000 constitute 9% of the sample, while those with an income below ₹7,80,000 - ₹10,00,000 make up the smallest group at 12%.
- These income distributions help us understand the socio-economic backgrounds of our respondents, which is crucial for analyzing how financial status might influence social media usage and its impact.

Social Media Usage

- Frequency of Use

Social Networking Usage Distribution



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Social Media Usage (Contd.)

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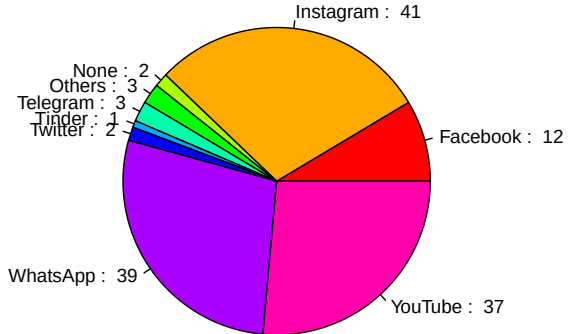
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- While collecting the data, we were also interested in knowing how often these people use social media/social networking sites. We're not surprised seeing that almost 16% of the population prefer to use social networking websites in a daily basis and they quite addicted to it. Although almost 36% and 34% of the people still prefer to use social networking sites frequently and moderate respectively by not adopting the new habits of these generation.

Social Media Usage (Preferred Platforms)

- Preferred Platforms

Favorite Social Networking Platform



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Social Media Usage (Preferred Platforms) (Contd.)

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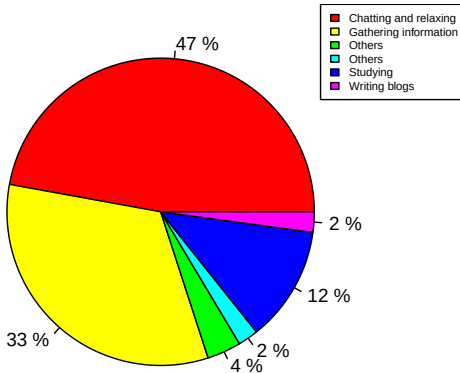
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- We collected the names of social networking sites/apps people used the most, and considered multiple names, and did a pie chart. Instagram emerged as the most used app, followed by Whats app and YouTube respectively.

Impact on Daily Life and Academic Performance

- Purpose of Using Social Networking Sites

Purpose of Using Social Networking Sites



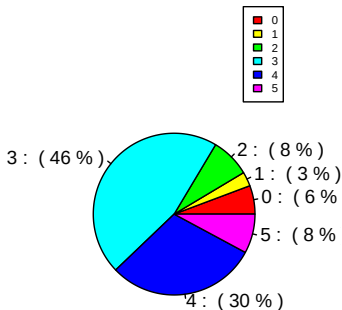
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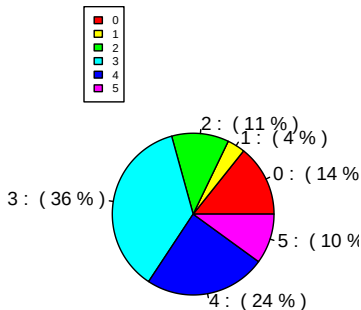
Academic and professional Satisfaction Rating

- since our data majorly focused on the teenager, there academic satisfaction level is a major thing when it comes to there daily life

Academic Satisfaction Rating



Professional Satisfaction Rating



The impact
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Academic and professional Satisfaction Rating (Contd.)

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- The majority of youth feel moderately satisfied in both overall and professional contexts.

Academic and professional Satisfaction Rating (Contd.)

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- The majority of youth feel moderately satisfied in both overall and professional contexts.
- There's a higher percentage of youth giving low satisfaction ratings (0-2) in professional satisfaction compared to overall satisfaction.

Academic and professional Satisfaction Rating (Contd.)

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- The majority of youth feel moderately satisfied in both overall and professional contexts.
- There's a higher percentage of youth giving low satisfaction ratings (0-2) in professional satisfaction compared to overall satisfaction.
- Positive Sentiment: A significant portion of the youth still rate their satisfaction highly (4-5) in both categories.

Behavioral Changes

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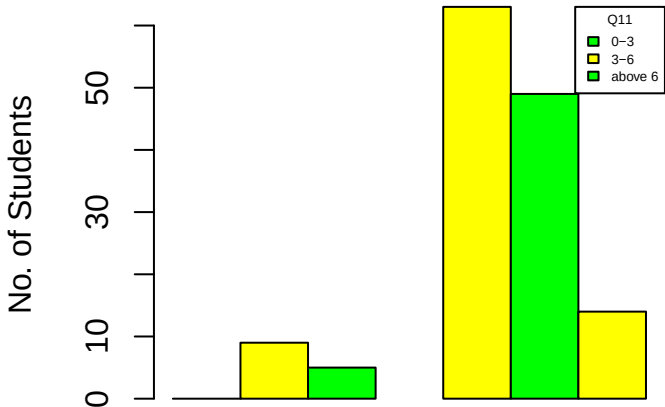
- Another interesting fact we can observe based on students reading behavior, we asked students whether they like to read or not? And we got some interesting results at the end, fact or the data showing us that there is no dependency when it comes to internet usage time of a student and reading behavior, most of the students like to read books although they spend many times on the internet.

Behavioral Changes (Contd.)

The impact of social networking sites on the students around Kolkata

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Swarnendu Dey, Roshni Majhi

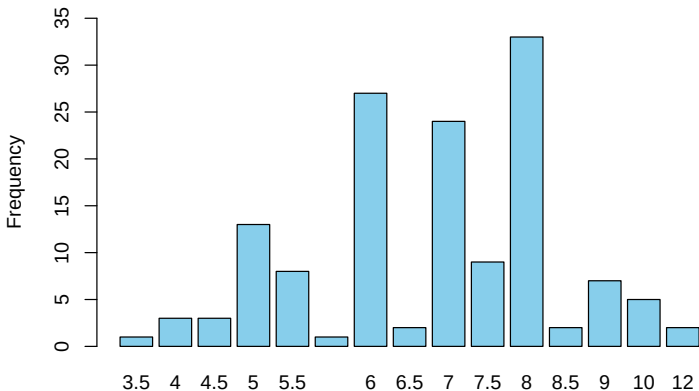
Internet Usage by Hours Spent on Internet



Behavioral Changes(Contd.)

- sleep pattern is a very important thing in our life ,not to mention the diseases caused by the lack of sleep.

Hours of Sleep in a Day



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Behavioral Changes(Contd.)

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- The data indicates that most individuals get between 6 to 8 hours of sleep, which aligns with general sleep recommendations for young adults.

Behavioral Changes(Contd.)

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- The data indicates that most individuals get between 6 to 8 hours of sleep, which aligns with general sleep recommendations for young adults.
- There are fewer individuals getting extremely low (less than 5 hours) or extremely high (more than 9 hours) amounts of sleep.

Behavioral Changes(Contd.)

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- The data indicates that most individuals get between 6 to 8 hours of sleep, which aligns with general sleep recommendations for young adults.
- There are fewer individuals getting extremely low (less than 5 hours) or extremely high (more than 9 hours) amounts of sleep.
- The peak at 8 hours suggests that a significant portion of the group is likely meeting the recommended amount of sleep, which is beneficial for health and well-being.

Behavioral Changes(Contd.)

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- The peak at 8 hours suggests that a significant portion of the group is likely meeting the recommended amount of sleep, which is beneficial for health and well-being.
- The histogram provides a clear visual representation of sleep patterns, showing that 6 to 8 hours of sleep is the most common duration among the group surveyed.

Behavioral Changes(Contd.)

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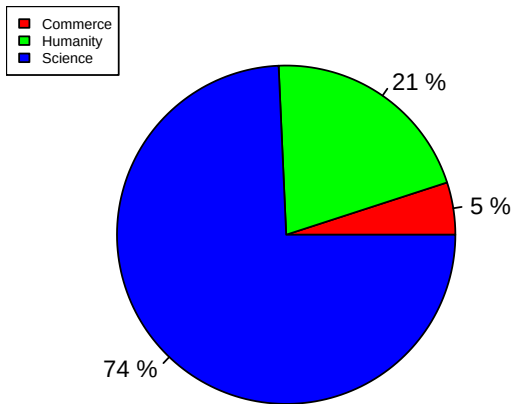
- The data indicates that most individuals get between 6 to 8 hours of sleep, which aligns with general sleep recommendations for young adults.
- There are fewer individuals getting extremely low (less than 5 hours) or extremely high (more than 9 hours) amounts of sleep.
- The peak at 8 hours suggests that a significant portion of the group is likely meeting the recommended amount of sleep, which is beneficial for health and well-being.
- The histogram provides a clear visual representation of sleep patterns, showing that 6 to 8 hours of sleep is the most common duration among the group surveyed.
- It is essential to encourage maintaining healthy sleep habits, aiming for 7-8 hours per night.

Visual Representations

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Stream of Study Distribution



Visual Representation (Contd.)

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- we can clearly see that the majority of the students belongs to science stream and in the other hand commerce students are rare to be found.

Visual Representation (Contd.)

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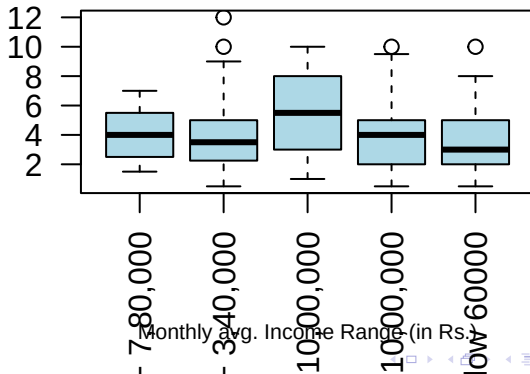
- we can clearly see that the majority of the students belongs to science stream and in the other hand commerce students are rare to be found.
- it's showing the increasing demand for science stream and also the unpopularity of commerce stream.

Contd.

Another visual representation is Distribution of Average Daily Expenditure of Time by Income Range

Avg. Daily Expenditure of Time (in hours)

Distribution of Average Daily Expenditure of Time by Income Range



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Contd.

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Observations:

- People with higher incomes tend to spend more time daily.

Contd.

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Observations:

- People with higher incomes tend to spend more time daily.
- Income groups with wider boxes have more variation in time use.

Contd.

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Observations:

- People with higher incomes tend to spend more time daily.
- Income groups with wider boxes have more variation in time use.
 - Outliers (unusual cases) can impact the overall pattern.

Contd.

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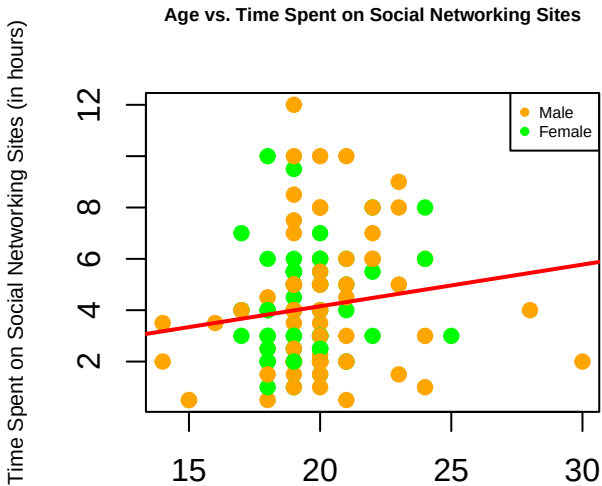
Observations:

- People with higher incomes tend to spend more time daily.
- Income groups with wider boxes have more variation in time use.
 - Outliers (unusual cases) can impact the overall pattern.
 - In summary, this graph helps us understand how income influences daily time allocation.

Regression Analysis

The impact of social networking sites on the students around Kolkata

Sourav Dey, Manish Chandra Roy, Swarnendu Dey, Roshni Majhi



Regression Analysis (Contd.)

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- We also tried to fit a regression to predict the average time spend on social networking sites given the age of the students, which although wasn't quite good.

Association check

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Swarnendu
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```
Loading required package:  grid
```

```
[1] 0.2428948
```

- There seems to be a weak to moderate association between age and time spent on social networking sites.

Inferential Data Analysis

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- For the categorical variables we have we decided to perform a chi-square test of independence.
- Hypothesis :

Inferential Data Analysis

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- Null Hypothesis (H_0): There is no relationship between the average number of hours spent on the internet and students' academic satisfaction levels.

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- Alternative Hypothesis (H_1): There is a relationship between the average number of hours spent on the internet and students' academic satisfaction levels.
- We divided the average internet usage into three groups: "0 – 3 hours", "3 – 6 hours", and "above 6 hours." The academic satisfaction levels were categorized from 0 to 5. Here is our contingency table:

Contd.

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Hours Spent on Internet	Academic Satisfaction (0-2)	Academic Satisfaction (3)	Academic Satisfaction (4-5)
0-3 hours	7	30	26
3-6 hours	8	29	21
Above 6 hours	8	5	6

Inferential Data Analysis (Contd.)

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	academic_satisfaction_level(0-5)		
avg_hour_they_spend_on_the_internet	0-2	3	4-5
0-3 hours	7	30	26
3-6 hours	8	29	21
above 6 hours	8	5	6

Pearson's Chi-squared test

data: cont_table

X-squared = 11.235, df = 4, p-value = 0.02405

Contd.

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Since, we see that the p-value of the chi-square test of independence is less than the level of significance (0.05) so in the light of the given data, we do not have sufficient reason to believe that the variables are not related, that is, we reject the null hypothesis, which states that avg. number of hours they spend on their internet does effect in their academic satisfaction level.

Inferential Data Analysis(Contd.)

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- Next, we tested the relationship between students' streams of study and their average time spent on social networking sites. The hypothesis was:

Inferential Data Analysis(Contd.)

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- Null Hypothesis (H_0): There is no relationship between the stream of study and average time spent on social networking sites.
- Alternative Hypothesis (H_1): There is a relationship between the stream of study and average time spent on social networking sites.

Contd.

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```
[1] "numeric"
```

```
[1] "character"
```

	0-3	3-6	above 6
SCI/THECH/MED	46	47	11
HUM/MBA/COMM/O.TW	17	11	8

Pearson's Chi-squared test

```
data: obs.freq
```

```
X-squared = 4.1084, df = 2, p-value = 0.1282
```

Contd.

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- Since, we see that the p-value of the chi-square test of independence is more than the level of significance (0.05) so in the light of the given data, we do have sufficient reason to believe that the variables are not related, that is, we accept the null hypothesis, which states that there is no relationship between a student's locality and his/her internet usage hours.

Inferential Data Analysis(Contd.)

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- We also explored the relationship between a student's locality and their internet usage hours. The hypothesis was:

Inferential Data Analysis(Contd.)

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Majhi

```
[1] "numeric"
```

```
[1] "character"
```

	0-3	above 3 hours
Surroundings	22	21
Kolkata	34	49
Villages	7	7

Pearson's Chi-squared test

```
data: obs.freq
```

```
X-squared = 1.3476, df = 2, p-value = 0.5098
```

Inferential Data Analysis(Contd.)

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Inferential Data Analysis(Contd.)

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- Lastly, we examined the relationship between the reason for internet usage and the stream of study. The hypothesis was:
- Hypothesis :

Inferential Data Analysis(Contd.)

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Inferential Data Analysis(Contd.)

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Inferential Data Analysis(Contd.)

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```
[1] "character"
```

```
[1] "character"
```

	SCI/THECH/MED	HUM/MBA/COMM/O.TW
Chatting and relaxing	51	15
Gathering information	34	12
studying	12	5

Pearson's Chi-squared test

```
data: obs.freq_Q15_Q10
```

```
X-squared = 0.3867, df = 2, p-value = 0.8242
```

Inferential Data Analysis(Contd.)

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Interpretations

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- These results imply that whether students are in the science stream or humanities stream, it does not significantly affect their pattern of internet usage. Students from both streams spend similar amounts of time on social networking sites and use the internet for similar reasons, such as chatting, gathering information, and studying.
- While these analyses show no significant relationship between stream of study and internet usage patterns, further research could explore other variables that might be influencing these patterns. Variables such as individual preferences, extracurricular activities, and academic workload could provide deeper insights.

Conclusions

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- In conclusion, our analysis reveals that being a science student or a humanities student does not significantly affect how students use the internet. Their patterns of internet usage, including time spent on social networking sites and reasons for internet use, are similar across different streams of study.

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THANK YOU!!